Hierarchical Employee Records and Tree Traversal

- Tree Structure Components 🌳
  - The organizational hierarchy is modeled as a tree structure starting from the root.
  - Parent nodes represent managers, while child nodes represent their direct reports.
  - Levels indicate the distance from the top-level executive or CEO.
  - Node and Level Organization
    - Each node stores specific employee data and pointers to subordinates.
    - Levels provide a clear visualization of the company's management depth.
    - Hierarchical Depth
      - Depth affects the time taken to reach employees at the bottom of the record.
- Traversal Techniques 🔍
  - Traversal is the process of visiting all nodes in the employee tree.
  - The choice of traversal impacts how quickly an individual can be located.
  - Depth-First Search (DFS)
    - Proceeds through a single management line to its lowest level before switching.
    - Uses a stack-based approach, either through recursion or an explicit stack.
  - Breadth-First Search (BFS)
    - Scans the hierarchy level by level, starting from the highest-ranking official.
    - Uses a queue-based approach to ensure all peers are checked before subordinates.
- Searching and Performance ⚡
  - Search efficiency is measured by the number of nodes visited during the process.
  - Performance depends heavily on whether the tree is balanced or skewed.
  - Efficiency Metrics
    - Both BFS and DFS have a time complexity of O(n) in the worst-case scenario.
    - Space complexity differs, with BFS requiring more memory for wide hierarchies.
  - Optimized Data Retrieval
    - BFS is the most suitable approach for retrieving employees close to the root.